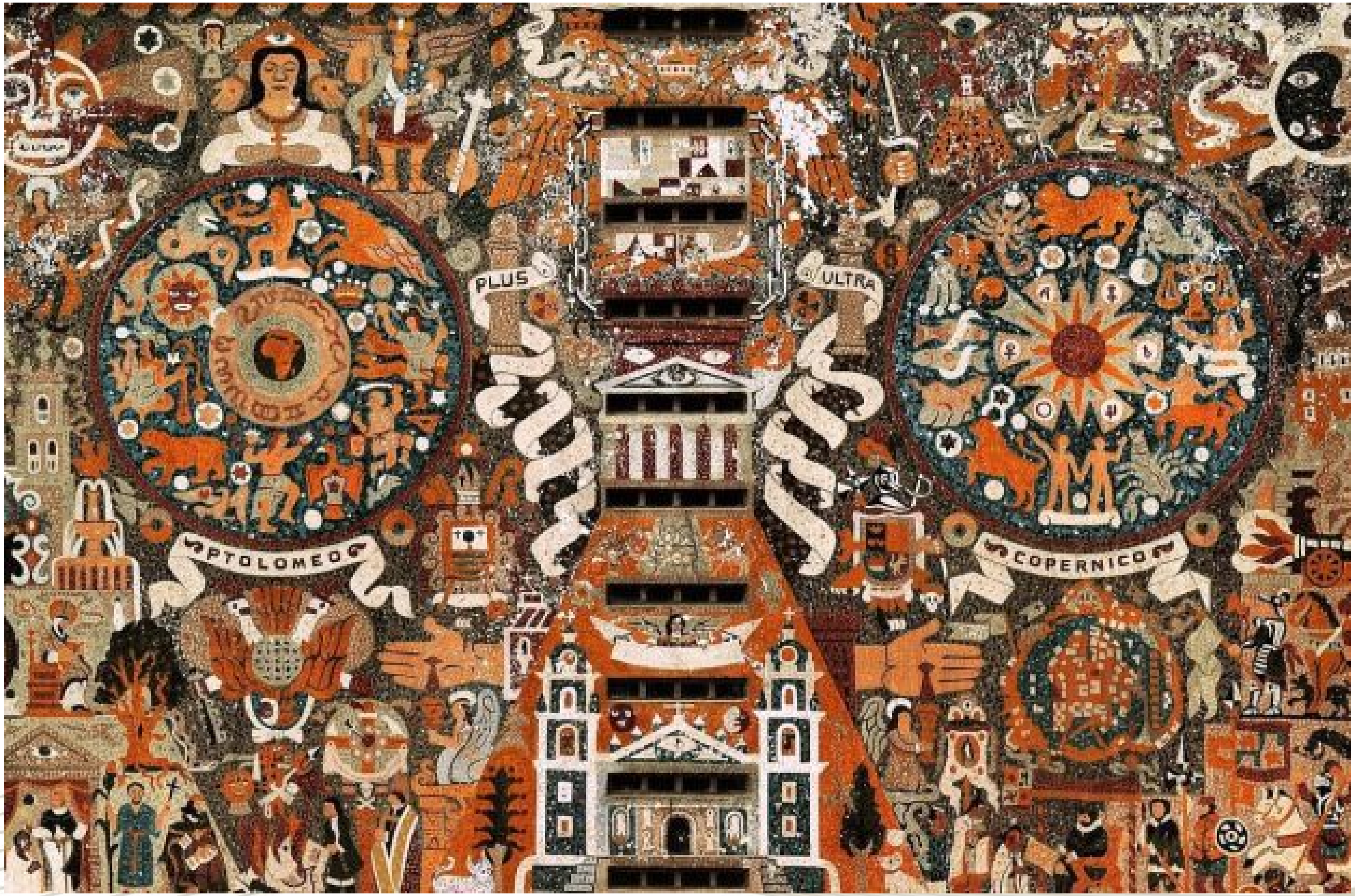
Evaluación de modelos

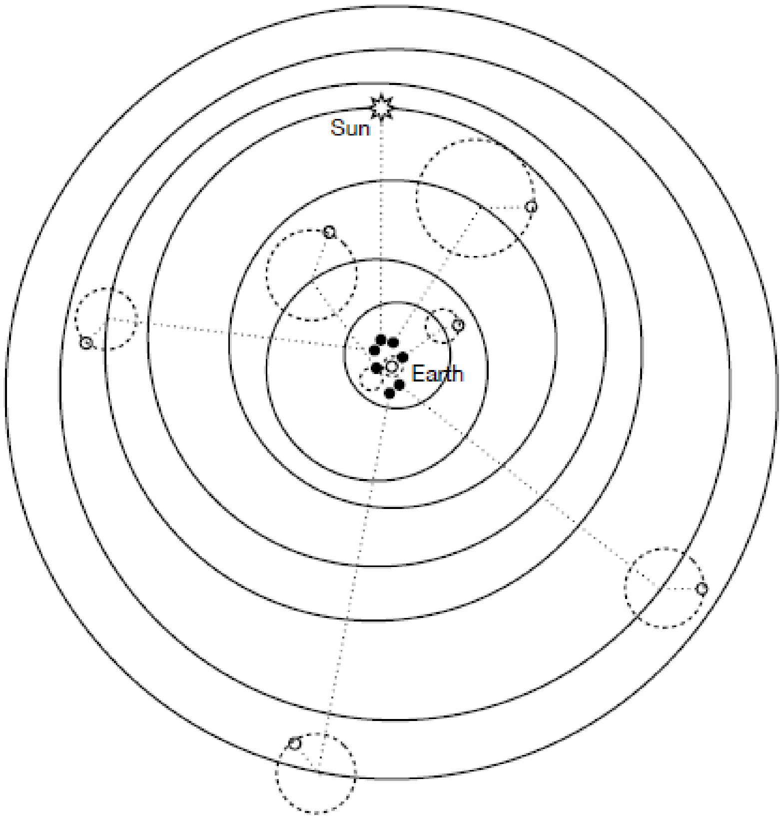
Dr. Héctor Nájera
Dr. Curtis Huffman

Accuracy v Simplicity

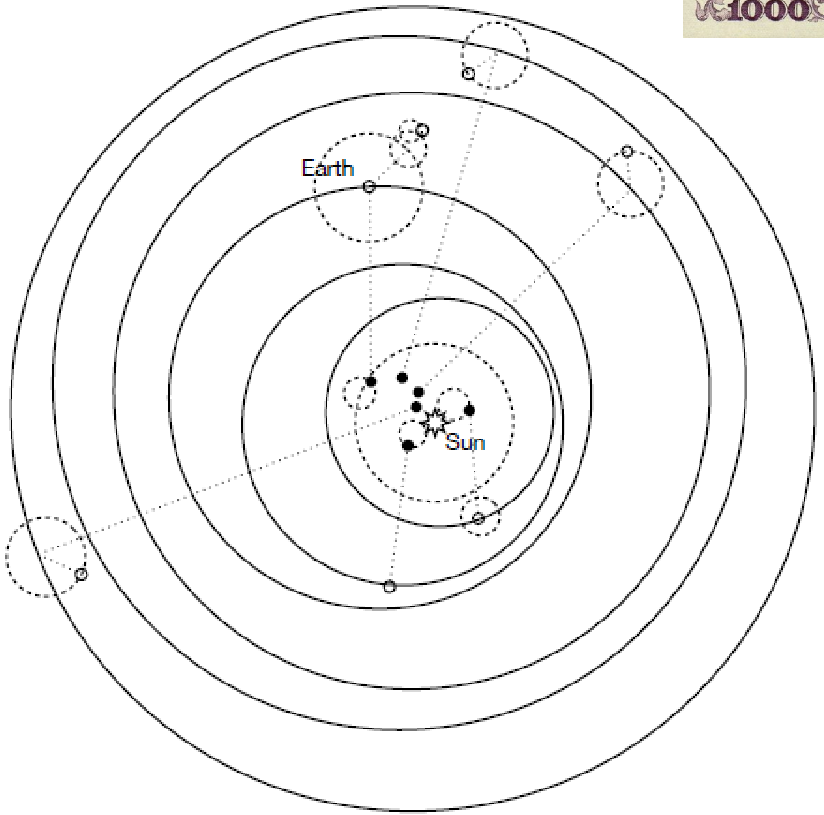


Accuracy v Simplicity

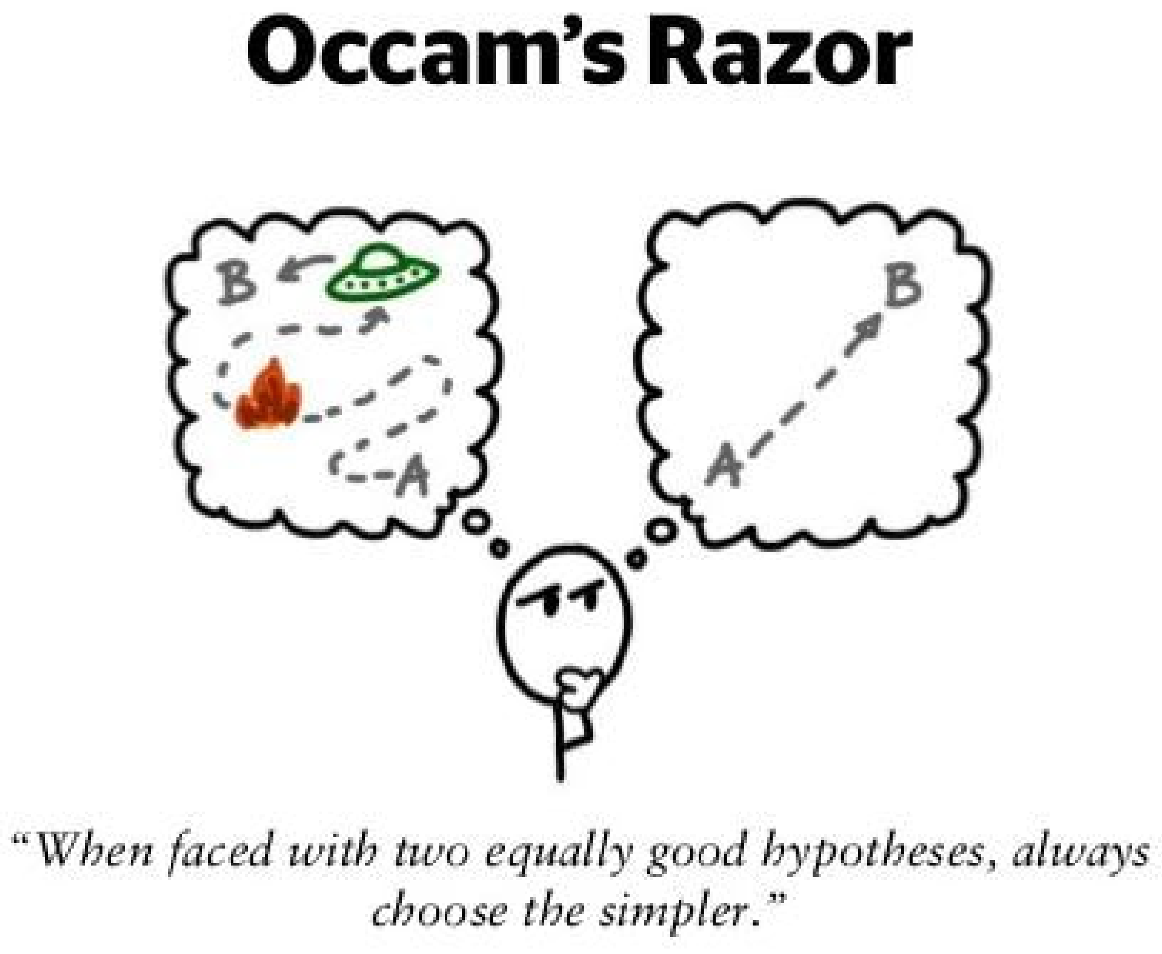
Ptolemaic Model



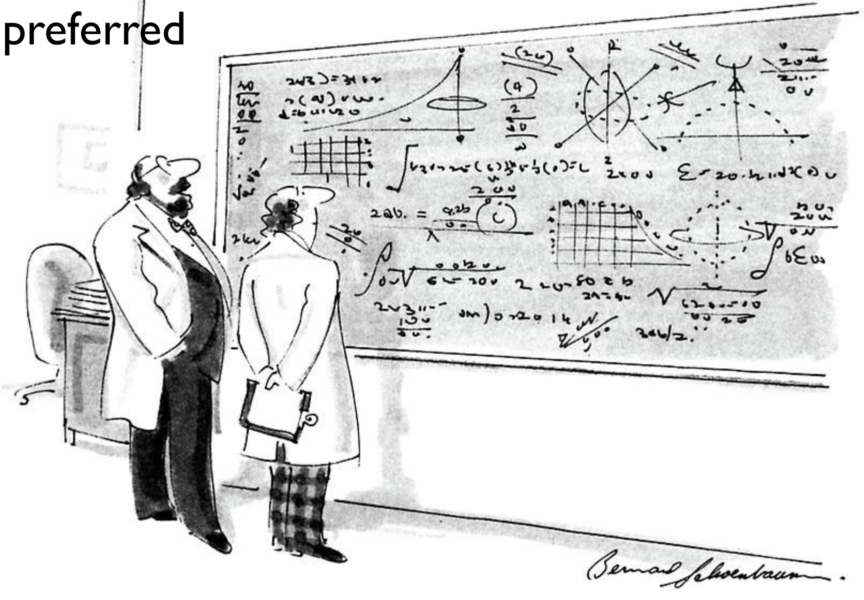
Copernican Model



Accuracy v Simplicity

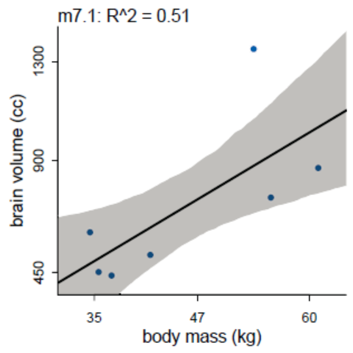


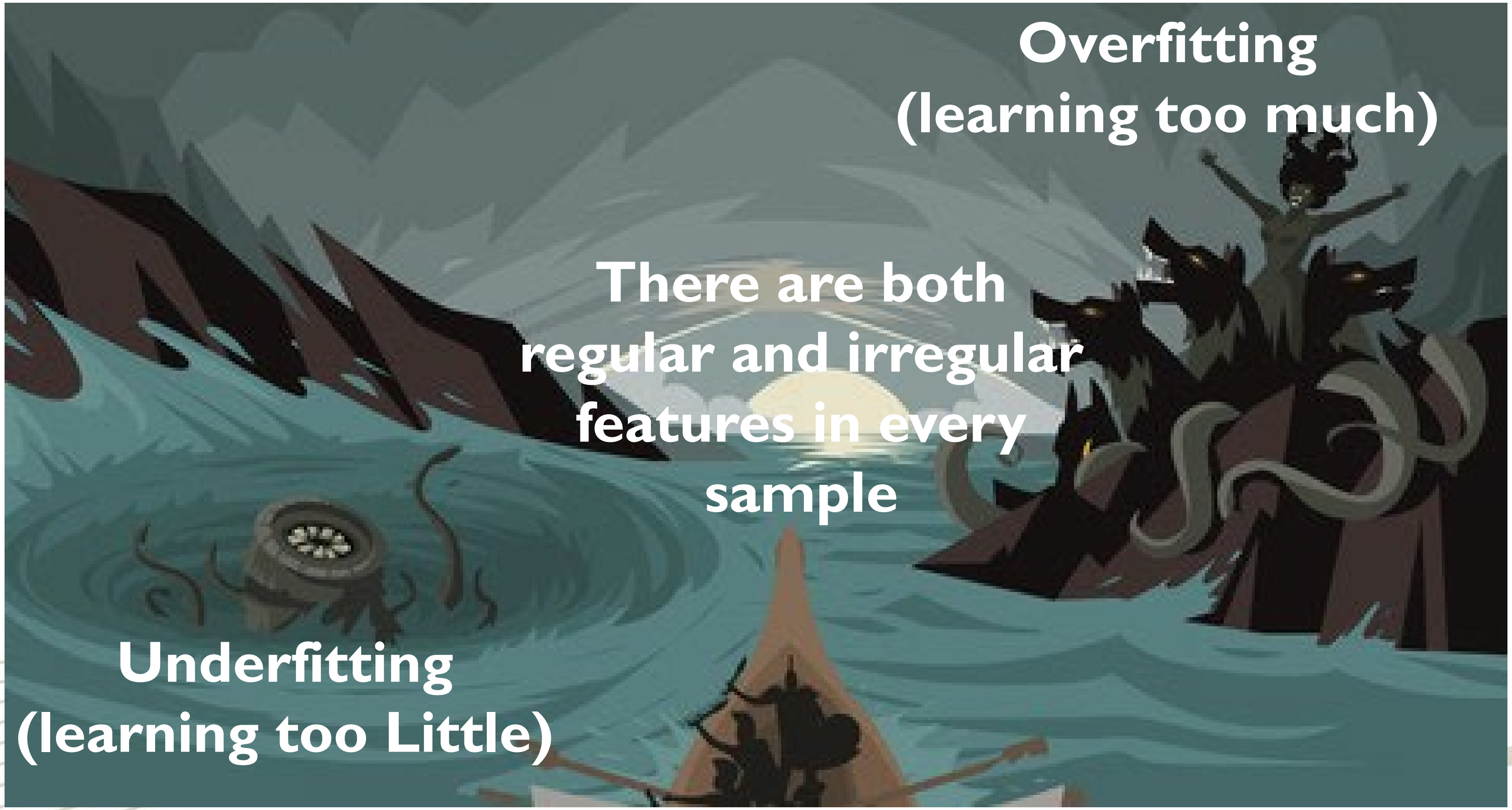
Models with fewer assumptions are to be preferred

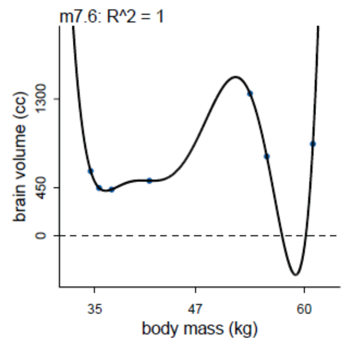

"Oh, if only it were so simple."

Accuracy v Simplicity

Making the model more complex (adding parameters) nearly always improves the fit of a model to the data (a measure of how well the model can retrodict the data used to fit the model), at the cost of predict new data worse.



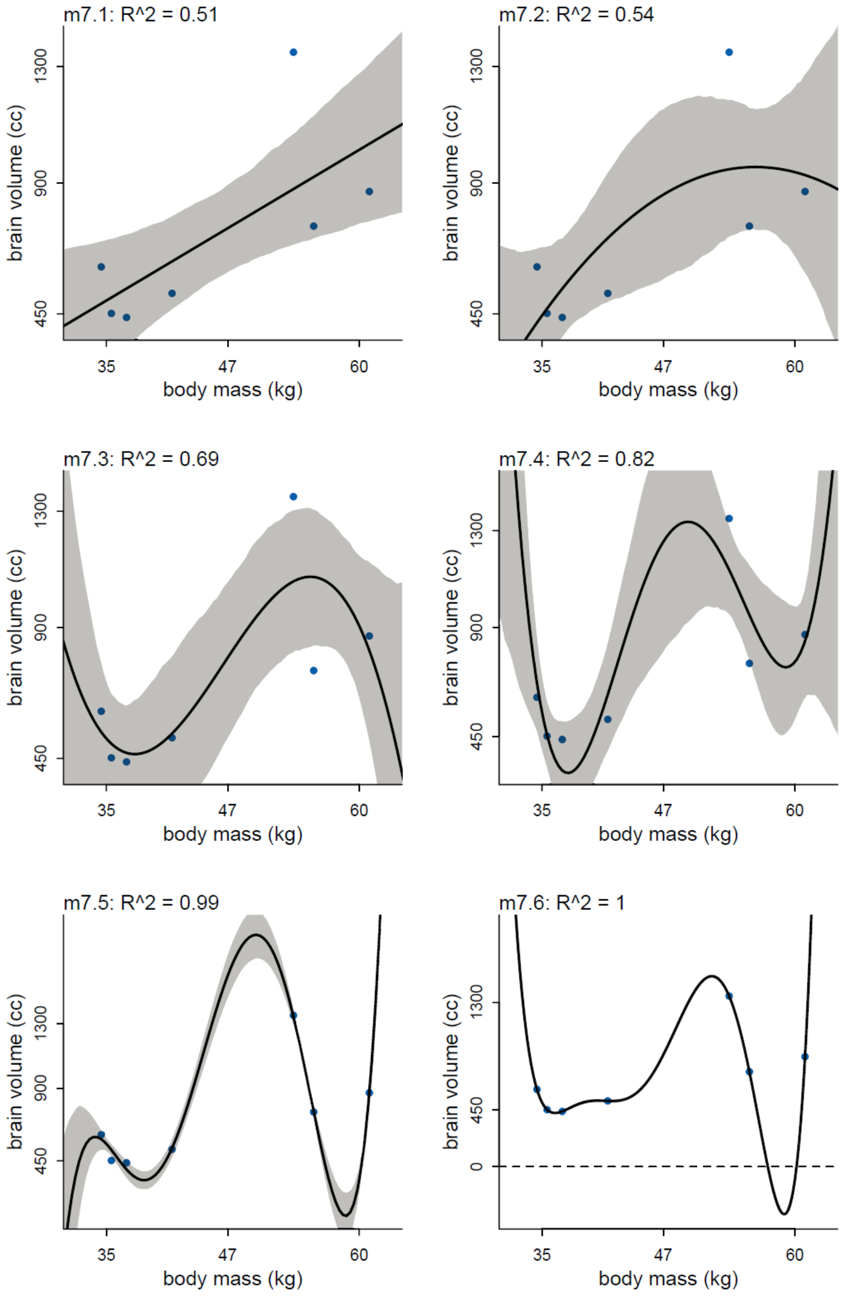




The proportion of variance "explained" by the model

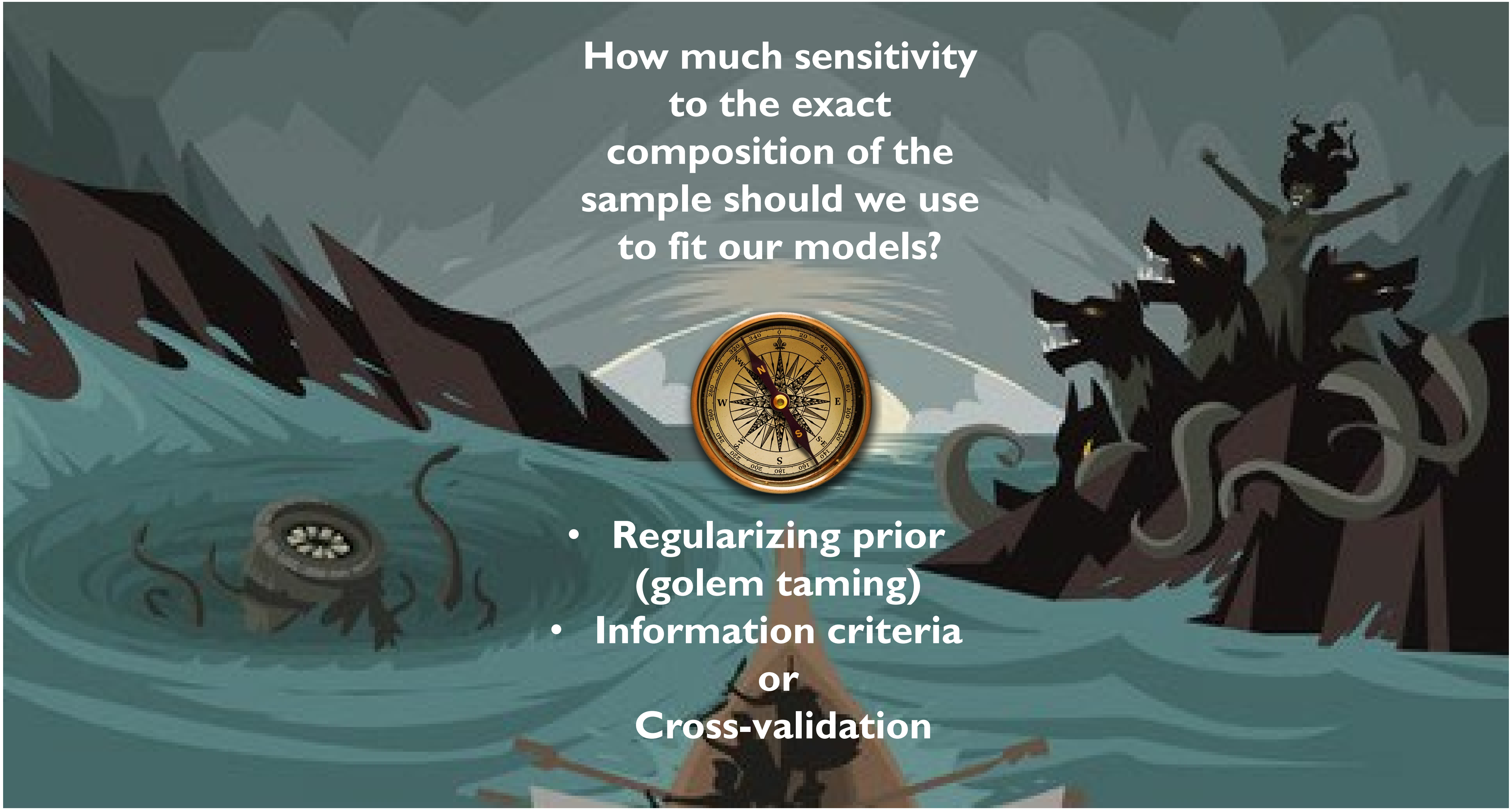
Too few parameters hurts, too

Stargazing doesn't help (***), the conventional 5% threshold is purely conventional, we shouldn't expect to optimize anything.



Encoding the sample is never the goal, we are not in the business of retrodiction. We don't care for irregularities.

The fit is perfect, but the model is ridiculus



How much sensitivity to the exact composition of the sample should we use to fit our models?

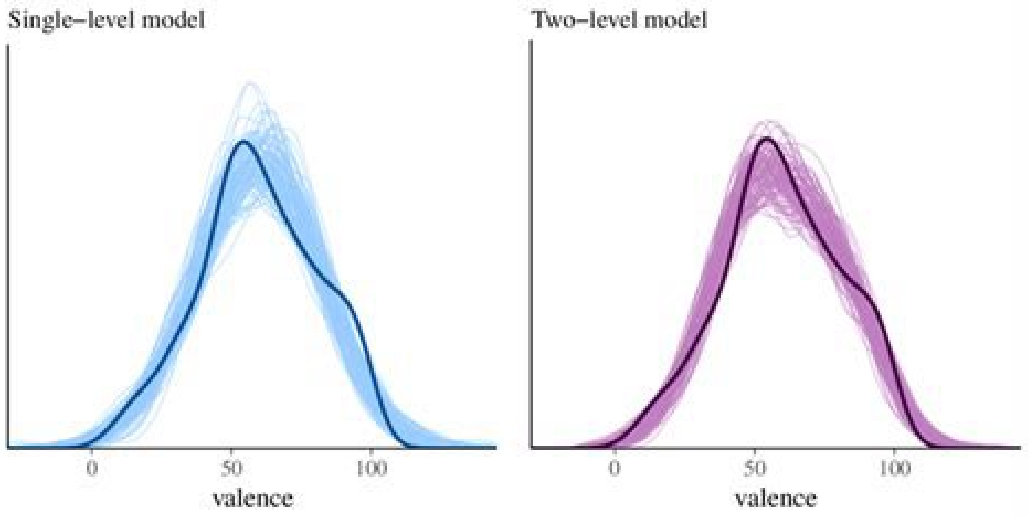
- **Regularizing prior (golem taming)**
- **Information criteria**

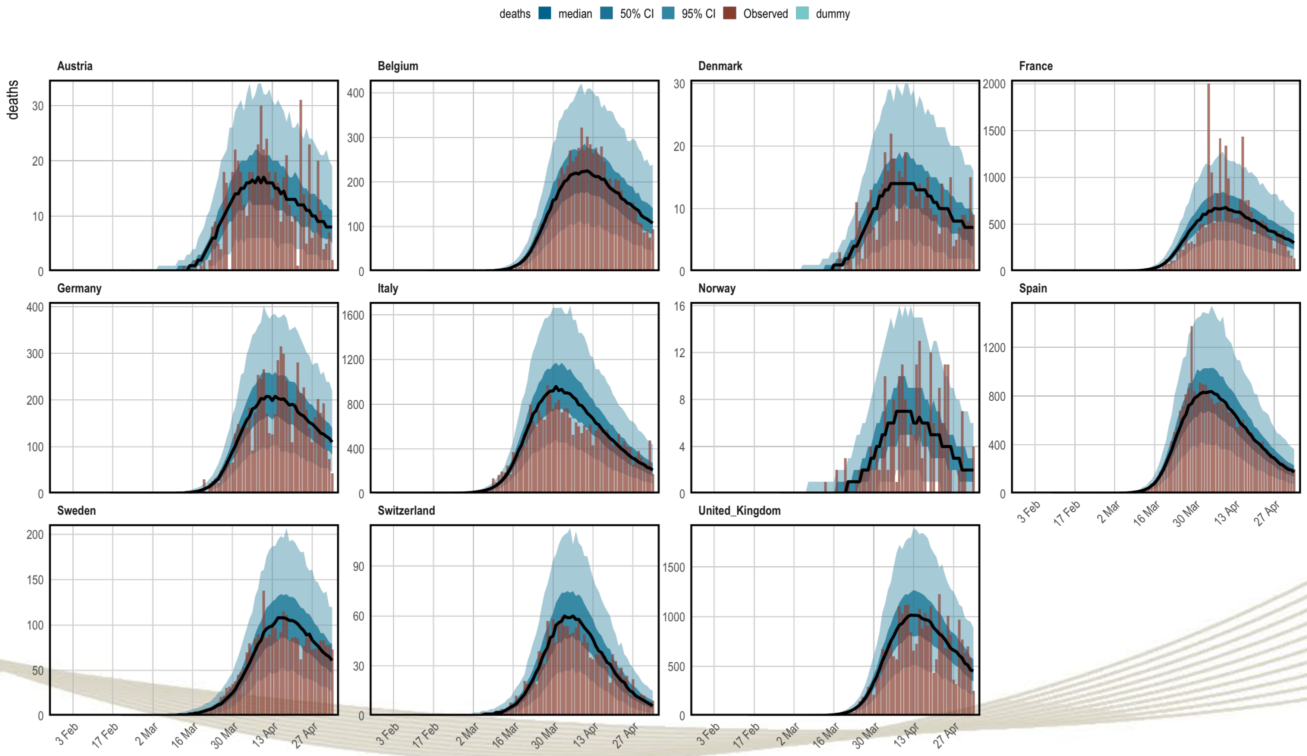
or

Cross-validation

Within-sample v Out-of-sample

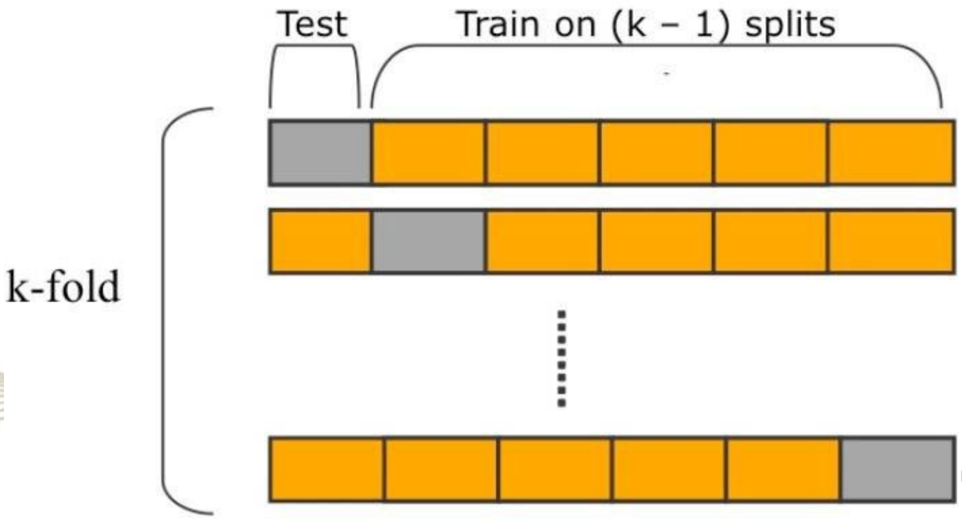
Predicción: Evaluación “formal” de un buen modelo (nunca within-sample)





Out-of-sample?

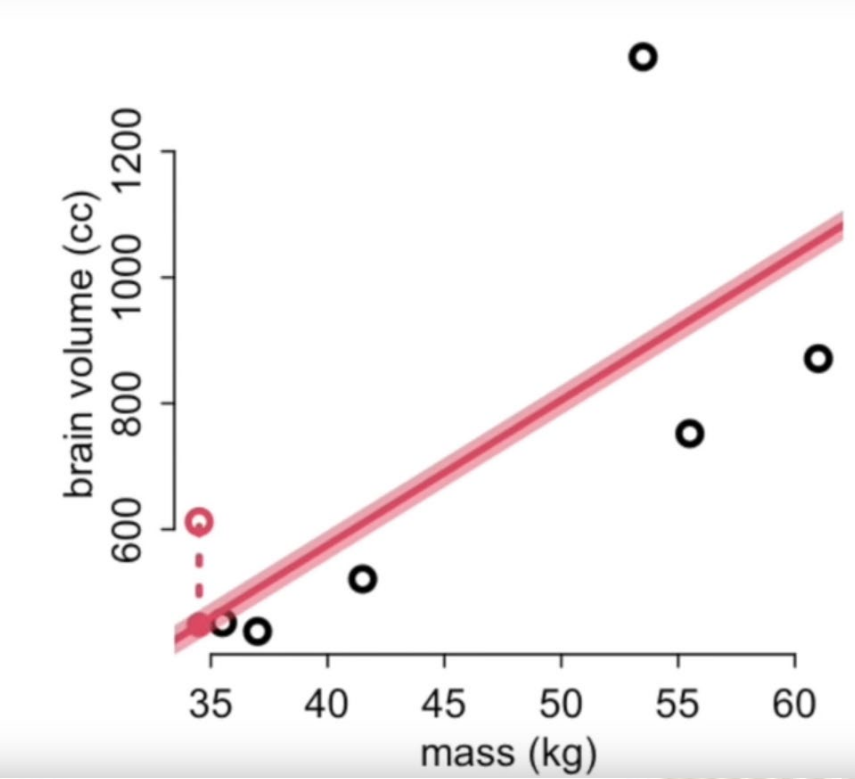
- How to evaluate your models out-of-sample?
  – Divide the sample in a number of chunks (folds). Predict each fold, after training on all the others.
    - How many chunks?
      – The maximum number of chunks (leave-one-out cross-validation)



Leave-one-out (loo) cross validation

¿Qué tan bueno es el modelo para predecir el siguiente punto –nuevas observaciones-?

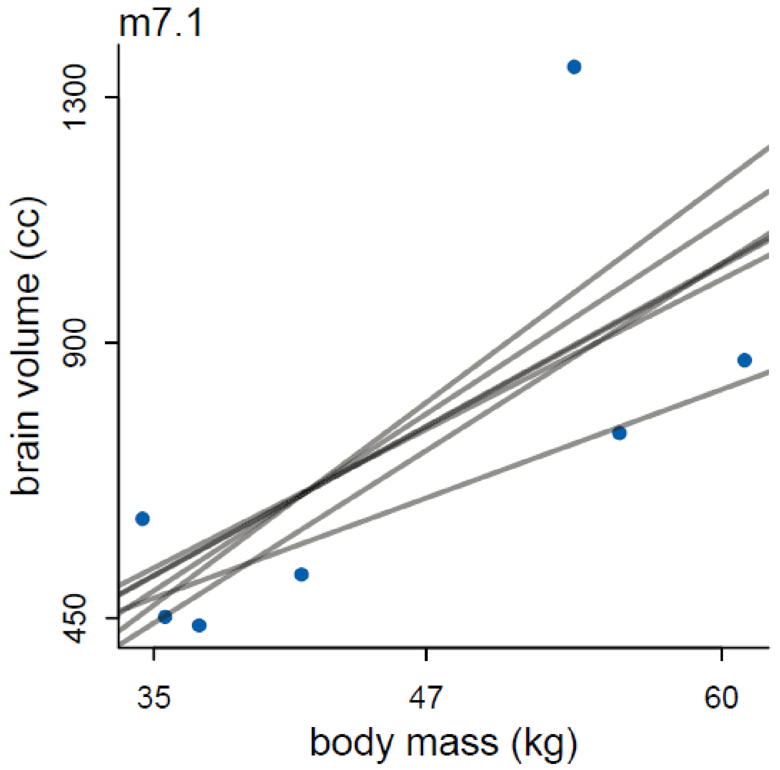
1. Tiramos un punto
2. Estimamos con el resto de la muestra
3. Predecimos el punto que tiramos -con 2-
4. Repetimos 1 con otro punto
5. Tomamos los errores de cada paso. MSE
   – Fuera
   – Dentro



Leave-one-out (loo) cross validation

¿Qué tan bueno es el modelo para predecir el siguiente punto –nuevas observaciones-?

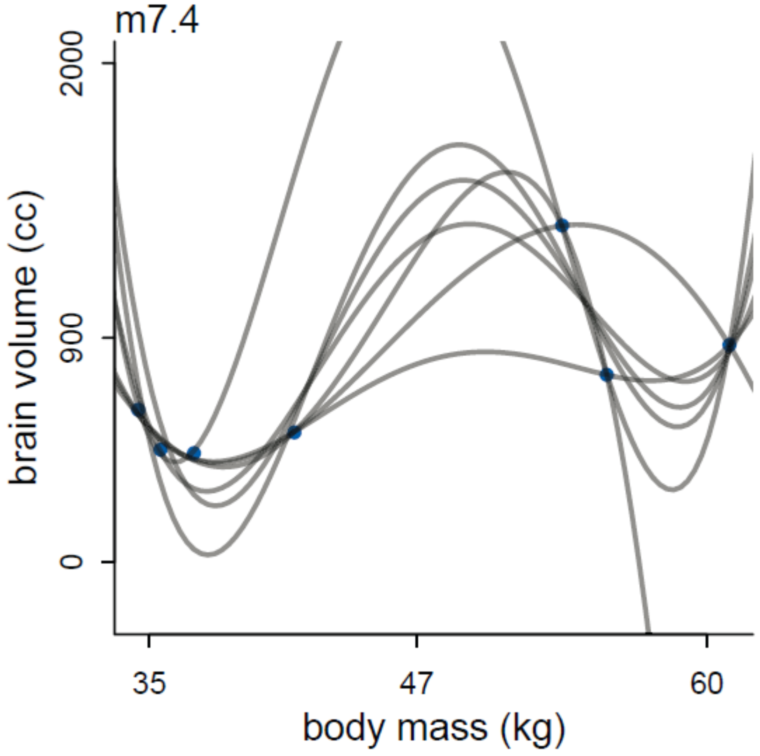
1. Tiramos un punto
2. Estimamos con el resto de la muestra
3. Predecimos el punto que tiramos -con 2-
4. Repetimos 1 con otro punto
5. Tomamos los errores de cada paso. MSE
   - Fuera
   - Dentro



Leave-one-out (loo) cross validation

¿Qué tan bueno es el modelo para predecir el siguiente punto –nuevas observaciones-?

1. Tiramos un punto
2. Estimamos con el resto de la muestra
3. Predecimos el punto que tiramos -con 2-
4. Repetimos 1 con otro punto
5. Tomamos los errores de cada paso. MSE
   – Fuera
   – Dentro



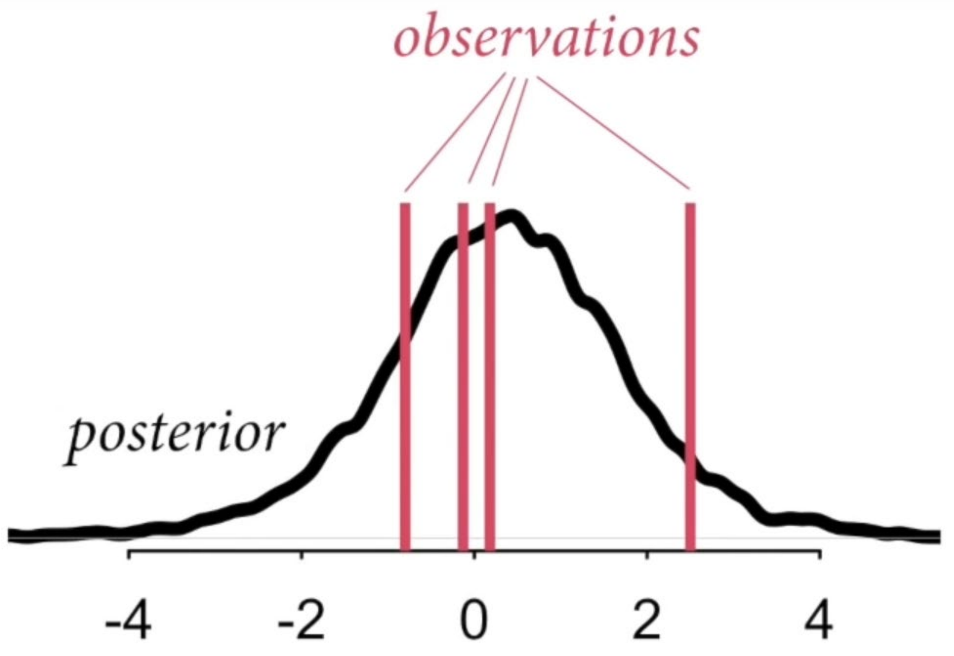
Bayesian loo and cross-validation

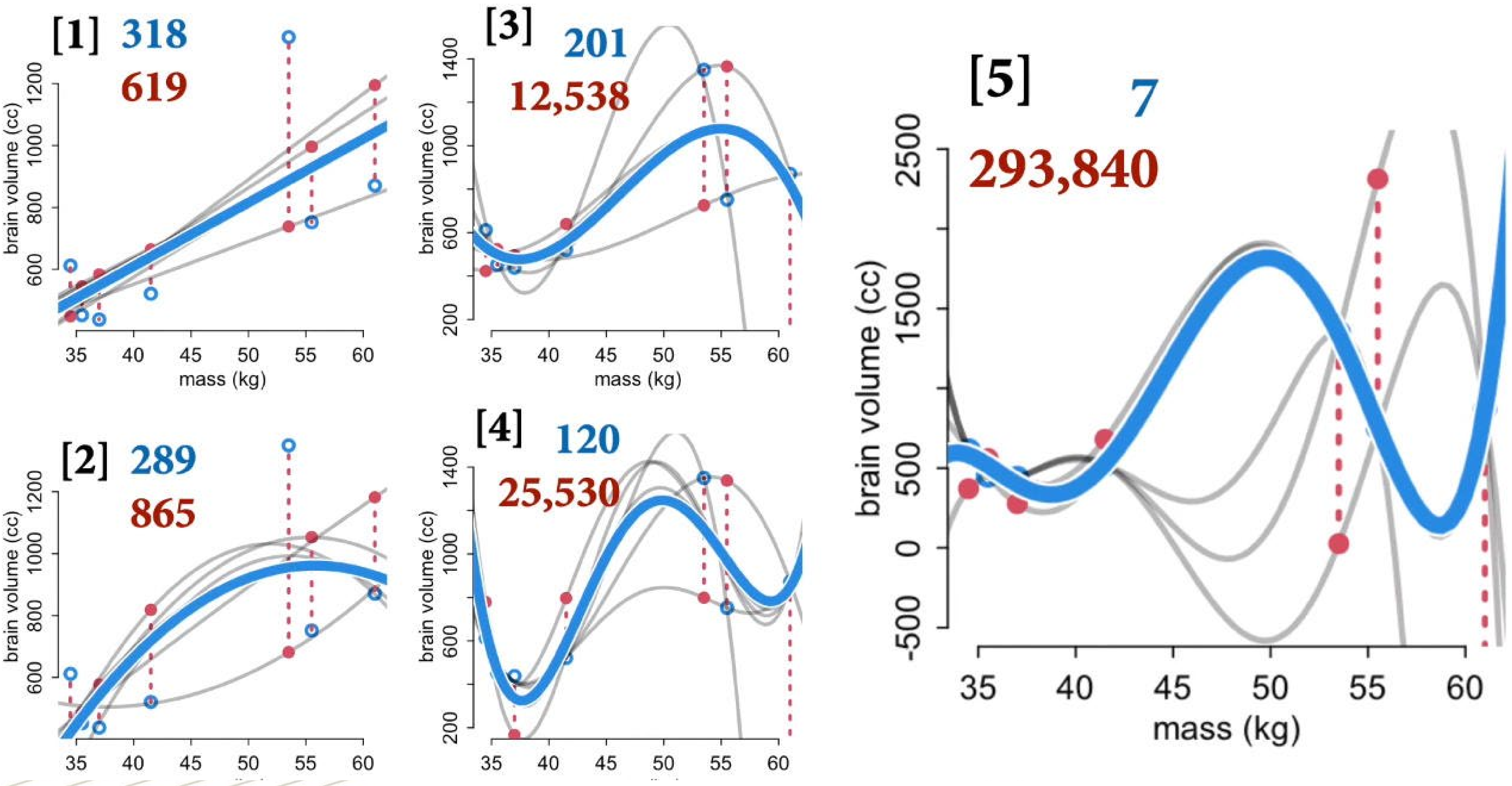
En Bayes no usamos puntos y una sola línea.

- Usamos toda la posterior, no punto a punto
- La probabilidad posterior de cada observación: la densidad predictiva de cada punto
- Obtenemos posteriores para cada conjunto menos un dato

¡¡El cómputo de esto es, para todo efecto práctico, imposible!!

Pero si lo fuera ¿Cómo luciría?





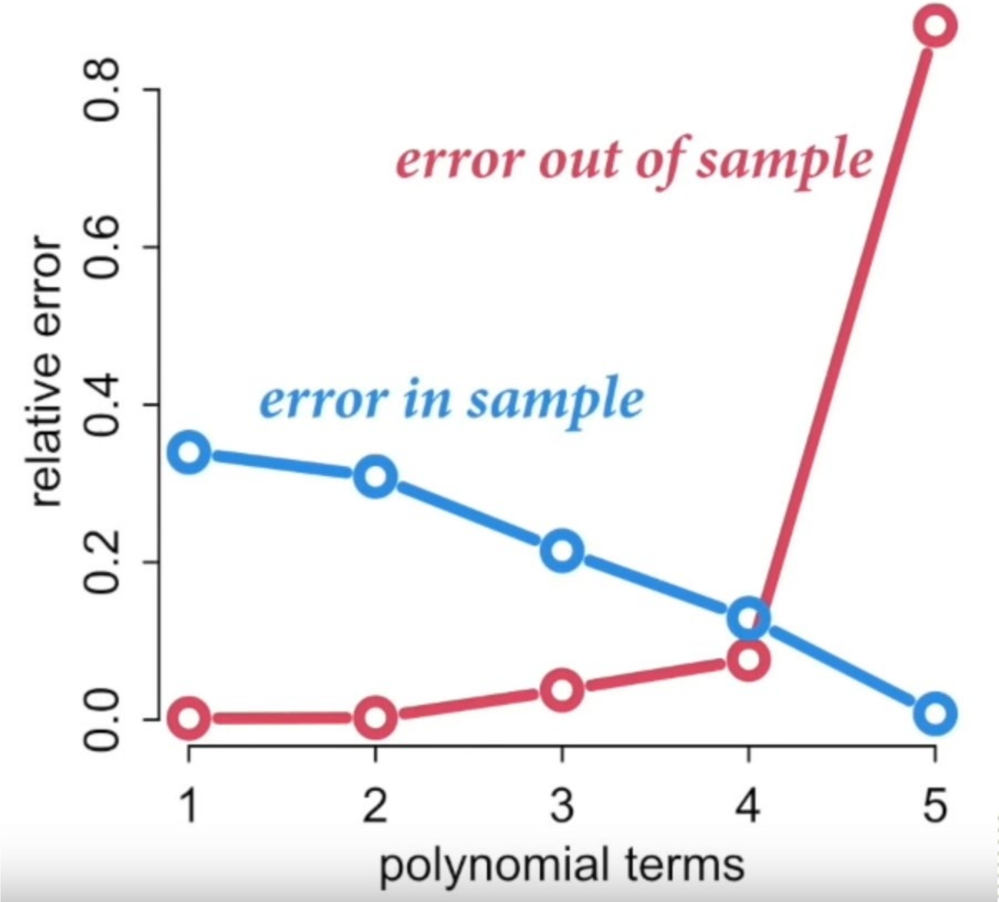
[1]
318
619
brain volume (cc)
600
800
1000
1200
35
40
45
50
55
60
mass (kg)
[2]
289
865
brain volume (cc)
600
800
1000
1200
35
40
45
50
55
60
[3]
201
12,538
brain volume (cc)
200
600
1000
1400
35
40
45
50
55
60
mass (kg)
[4]
120
25,530
brain volume (cc)
200
600
1000
1400
35
40
45
50
55
60
[5]
7
293,840
brain volume (cc)
-500
0
500
1500
2500
35
40
45
50
55
60
mass (kg)

Cross-validation

Comparo distintos modelos candidatos

- Error in sample > error out sample
- A medida que el modelo es más complejo, su poder predictivo fuera de la muestrea cae

Flexibilidad ~ ¡¡sobreajuste!!

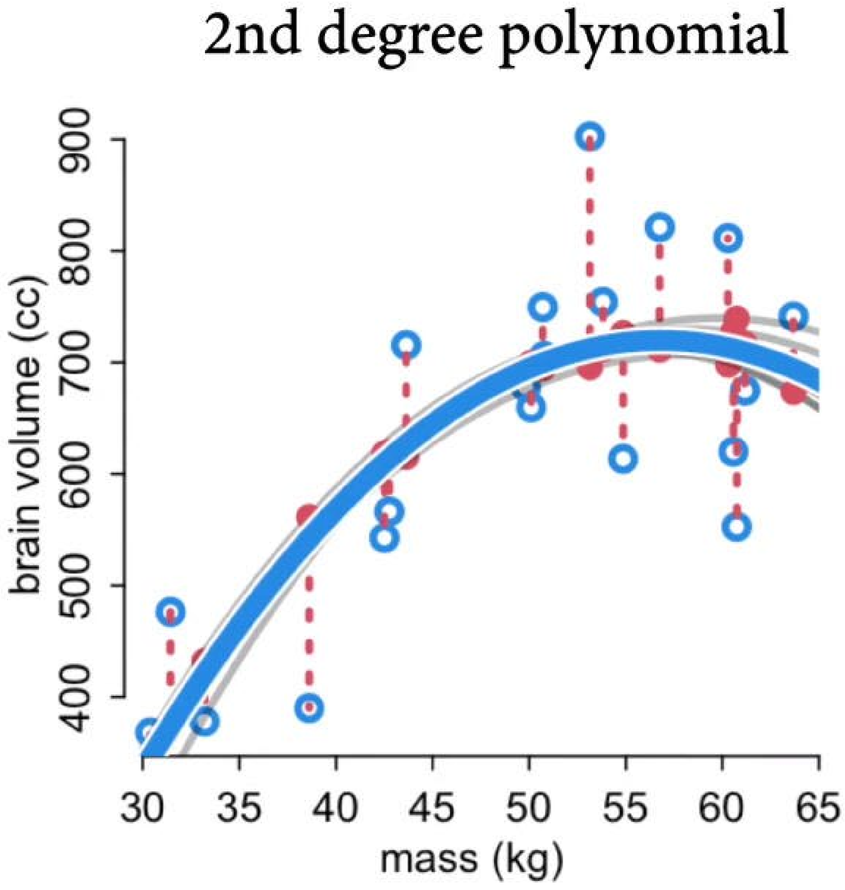


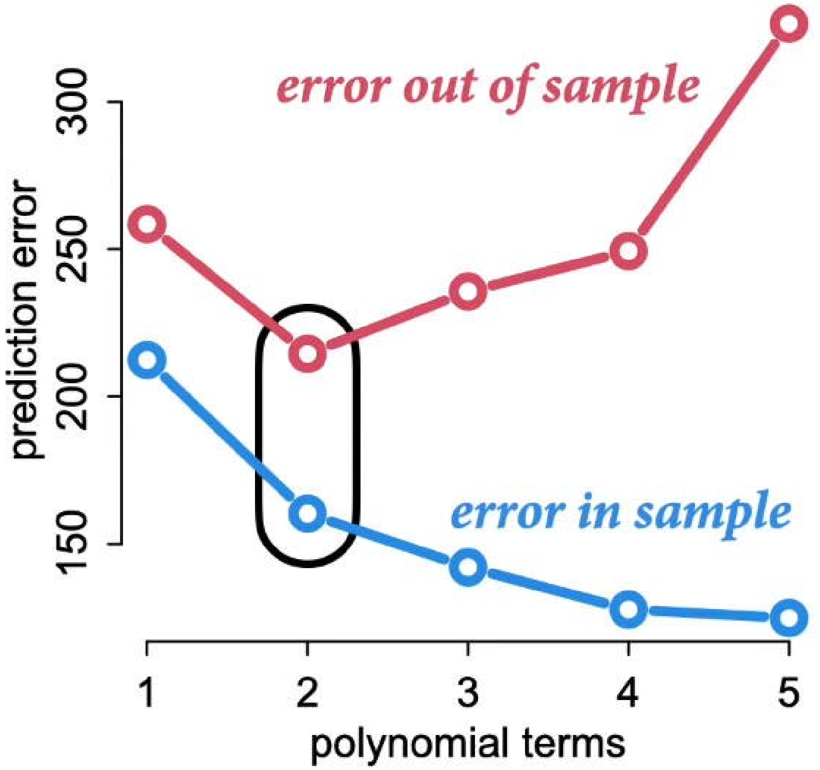
Cross-validation: fitting is easy, but prediction is hard

Validación cruzada es una medida de sobreajuste

Regularizacion: función que encuentra aspectos regulares del proceso

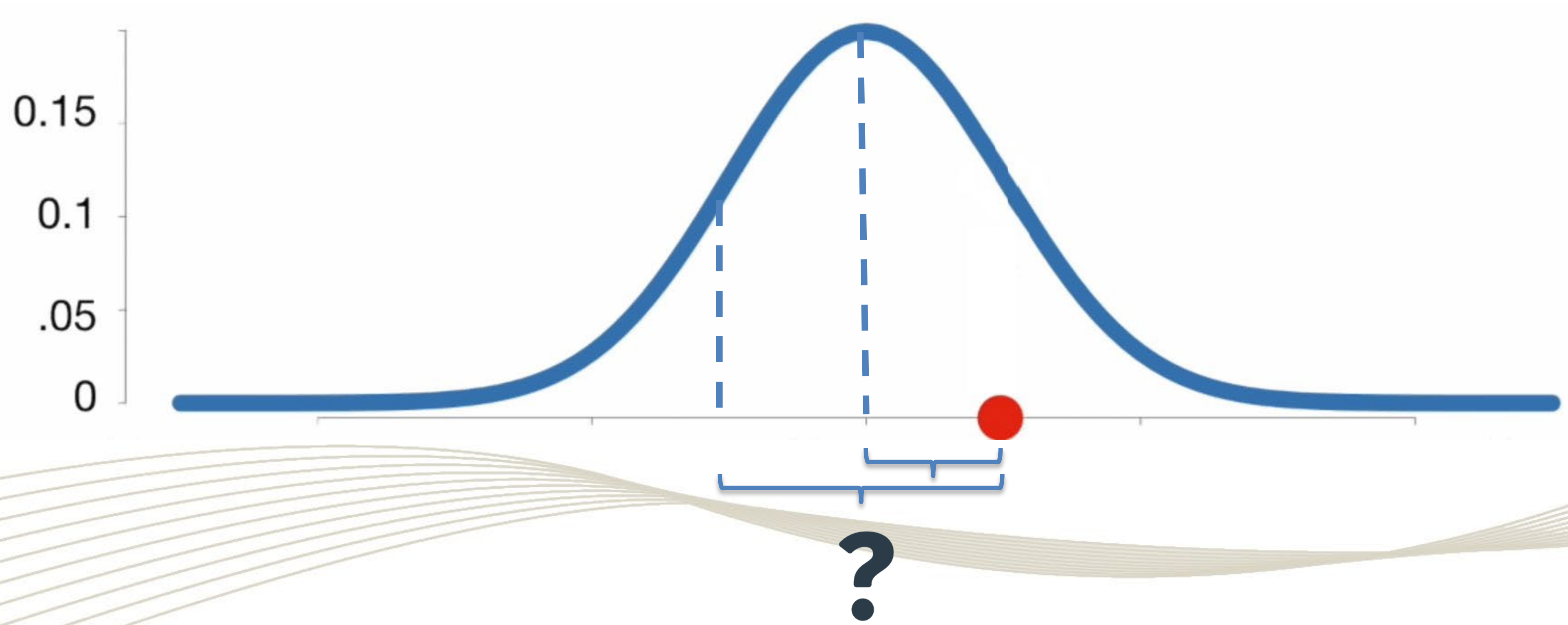
Priors, priors, priors… reducir la flexibilidad del modelo





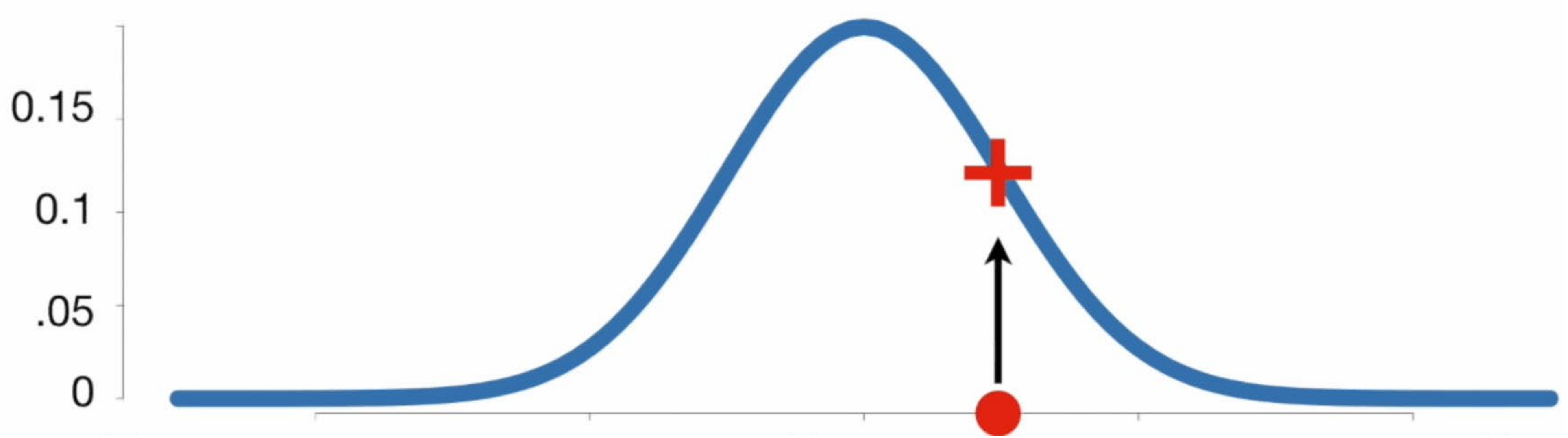
Somos bayesianos en este curso

- No usamos puntos, usamos distribuciones
  - Las estimaciones son distribuciones, los puntos decisiones.
  - No tomamos decisiones en este curso, comunicamos nuestras estimaciones y con base en ellas deliberamos en comunidad.



Somos bayesianos en este curso

- No usamos puntos, usamos distribuciones
  - Las estimaciones son distribuciones, los puntos decisiones.
  - No tomamos decisiones en este curso, comunicamos nuestras estimaciones y con base en ellas deliberamos en comunidad.



Bayesian Cross-Validation

S samples from posterior

N data points

log pointwise predictive density

$$\text{lppd}_{\text{CV}} = \sum_{i=1}^{N} \frac{1}{S} \sum_{s=1}^{S} \log \Pr(y_i | \theta_{-i,s})$$

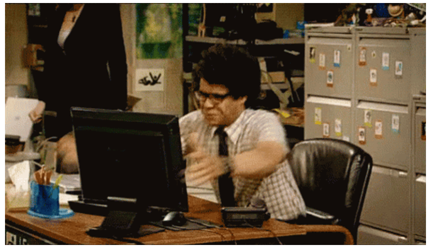
average log probability for point i

log probability of each point i, computed with posterior that omits point i

¡Usamos toda la distribución posterior, no sólo la predicción puntual!

LOO-CV does not scale well to large datasets

¡¡El cómputo de esto es, para todo efecto práctico, imposible!!

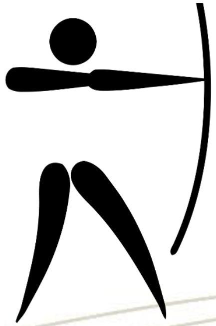


- LOO-CV es demasiado costosa e ineficiente
- ¿Qué tal si podemos calcular la brecha a partir de una sola estimación?
  - **Information criteria (WAIC): Widely applicable information criteria**. Teoría de la información
  - **Importance sampling (PSIS): Pareto smooth importance sampling.**
    - Point with low probability has a strong influence on the posterior distribution

Using tools like PSIS and WAIC is much easier than understanding them. Which makes them quite dangerous.

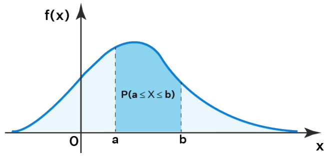
Measuring accuracy

- What do you want the model to do well at?
- What do we talk about when we talk about accuracy?

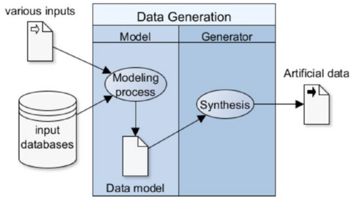


Point

Distribution



Model generating the data

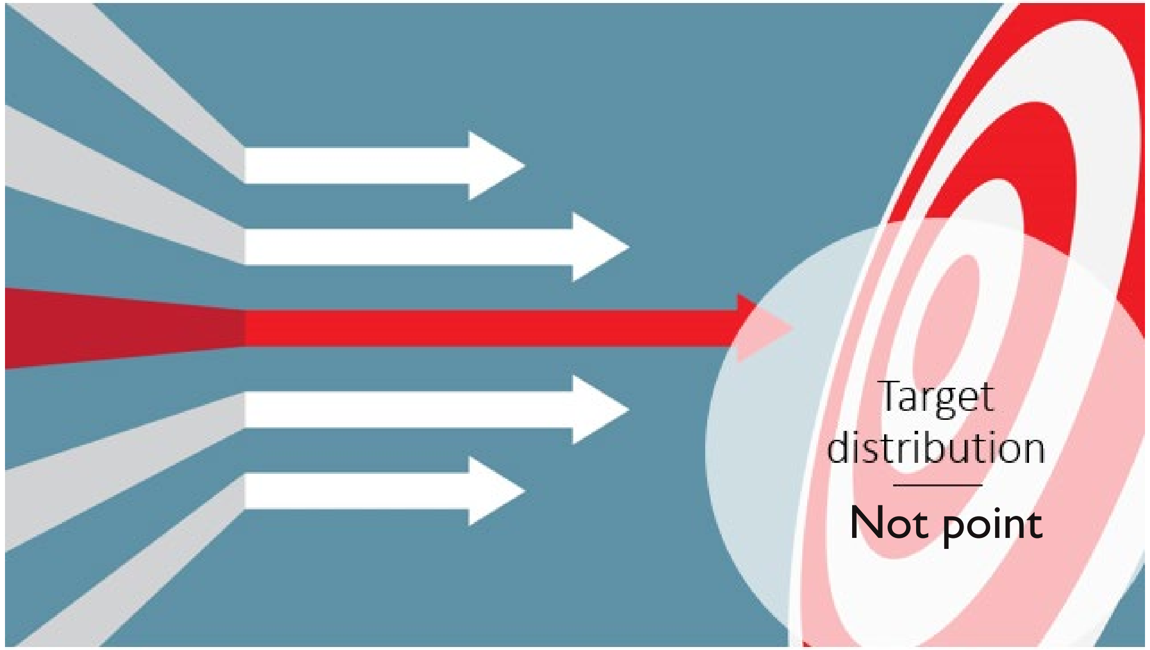


Measuring accuracy (Information Theory)

- Which definition of accuracy is maximized by knowing the true model generating the data?

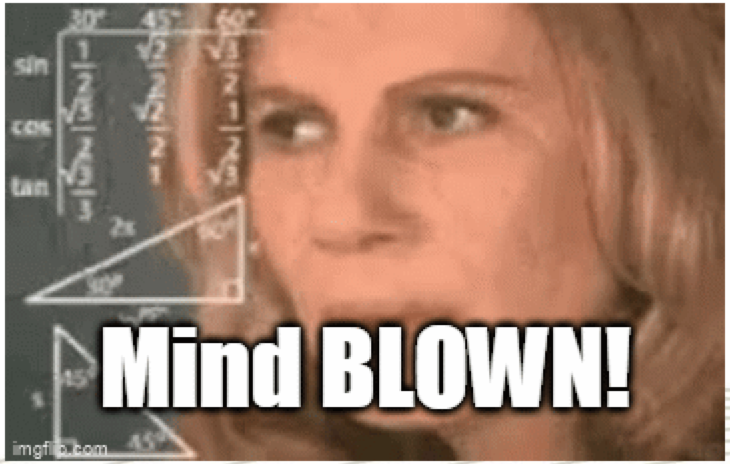
  The joint likelihood $P(D|\theta)$

- Information theory: a measurement scale for distance from perfect accuracy (out of sample deviance)



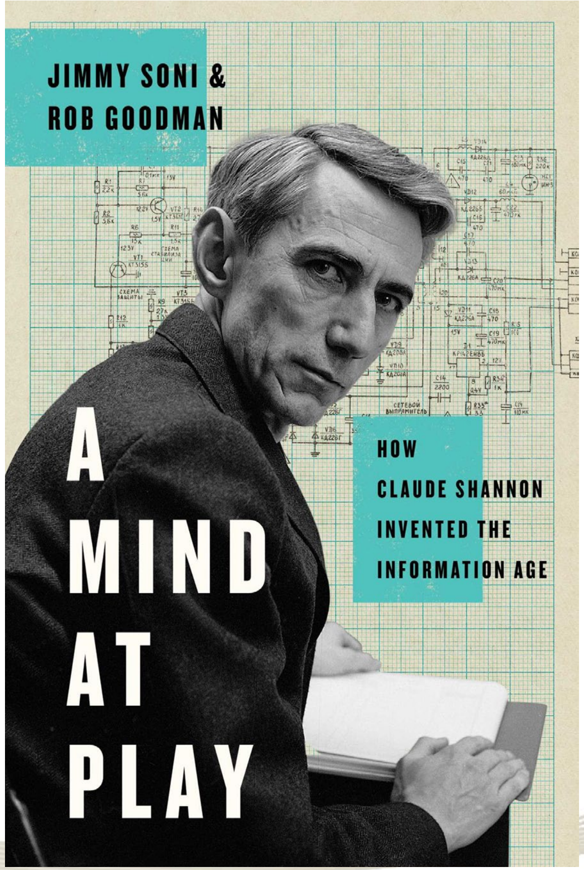
Measuring accuracy (Information Theory)

- **The true data-generating model will not have the highest hit rate**



Information criteria

- Implementing them is much easier tan understanding them (Information Theory)
  - The bit player (https://youtu.be/JP1Ljp8X6hg)



Validación cruzada y criterios de información

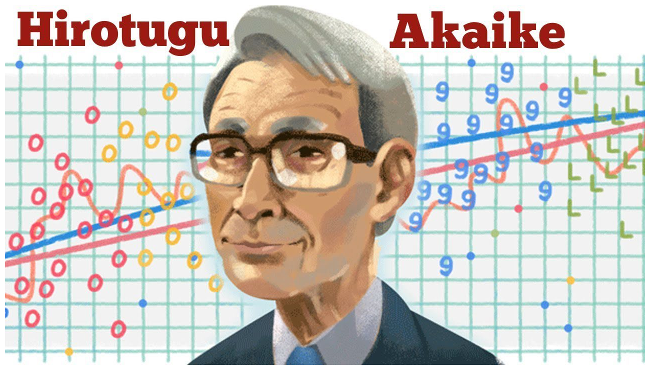
PREDICIENDO EL DESEMPEÑO PREDICTIVO

Information criteria

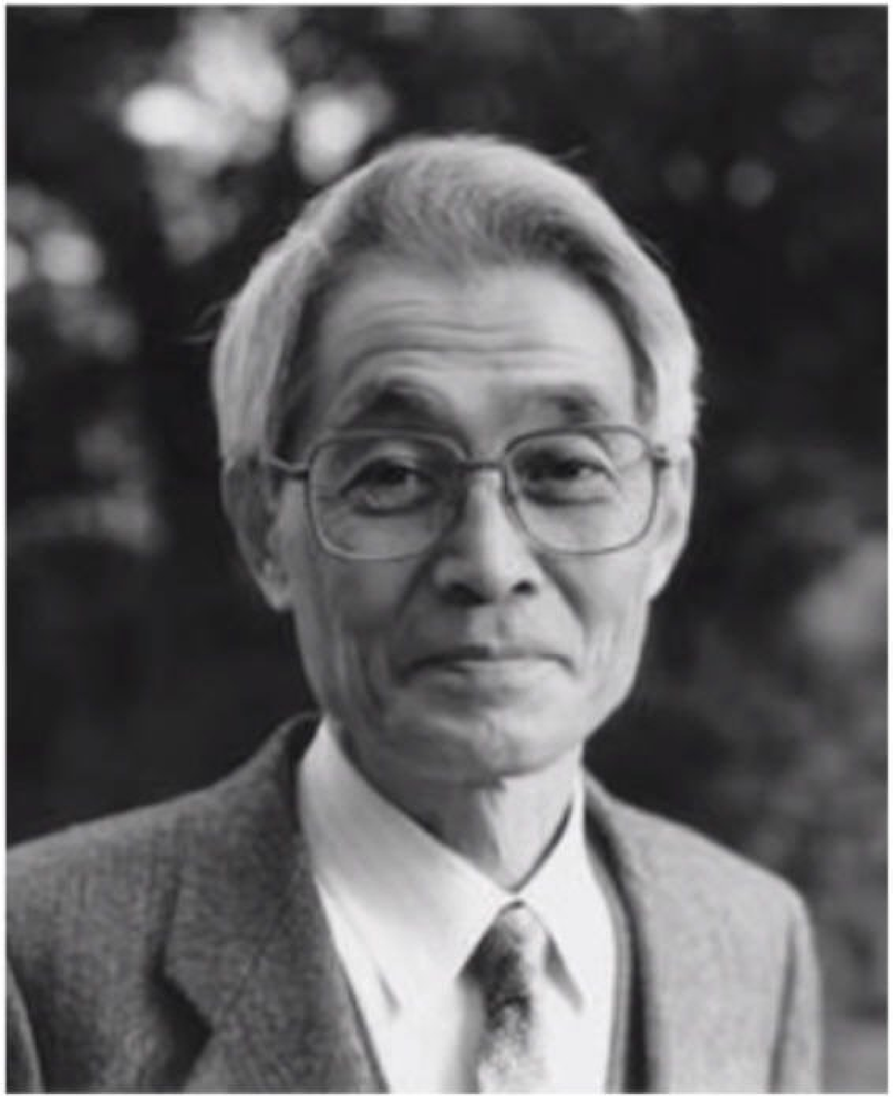
- Constructs a theoretical estimate of the relative out-of-sample KL divergence
  - The difference between training deviance and testing deviance is about twice the number of parameters

$$AIC = D_{train} + 2p = -2lppd + 2p$$

The dimensionality of the posterior distribution is a natural measure of the model's overfitting tendency



"On the morning of March 16, 1971, while taking a seat in a commuter train, I suddenly realized that the parameters of the factor analysis model were estimated by maximizing the likelihood and that the mean value of the logarithmus of the likelihood was connected with the Kullback-Leibler information number."



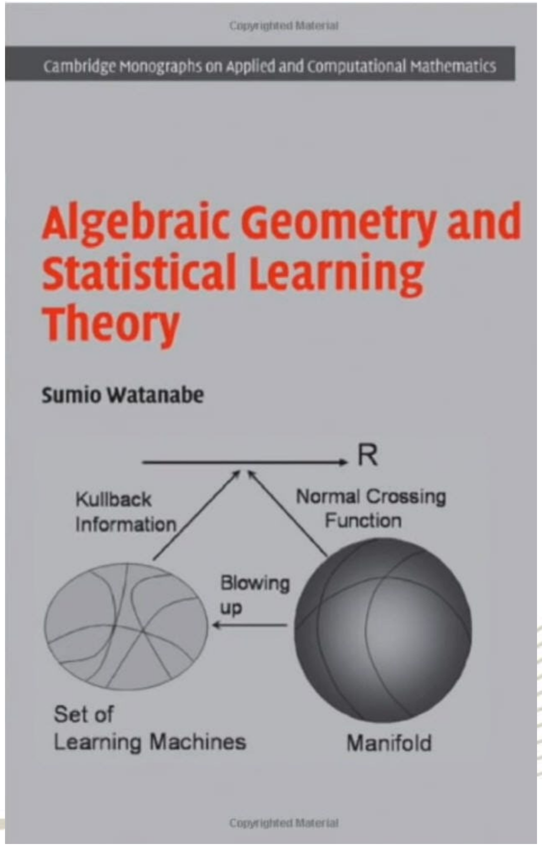
Hirotugu Akaike (1927–2009)

赤池弘次

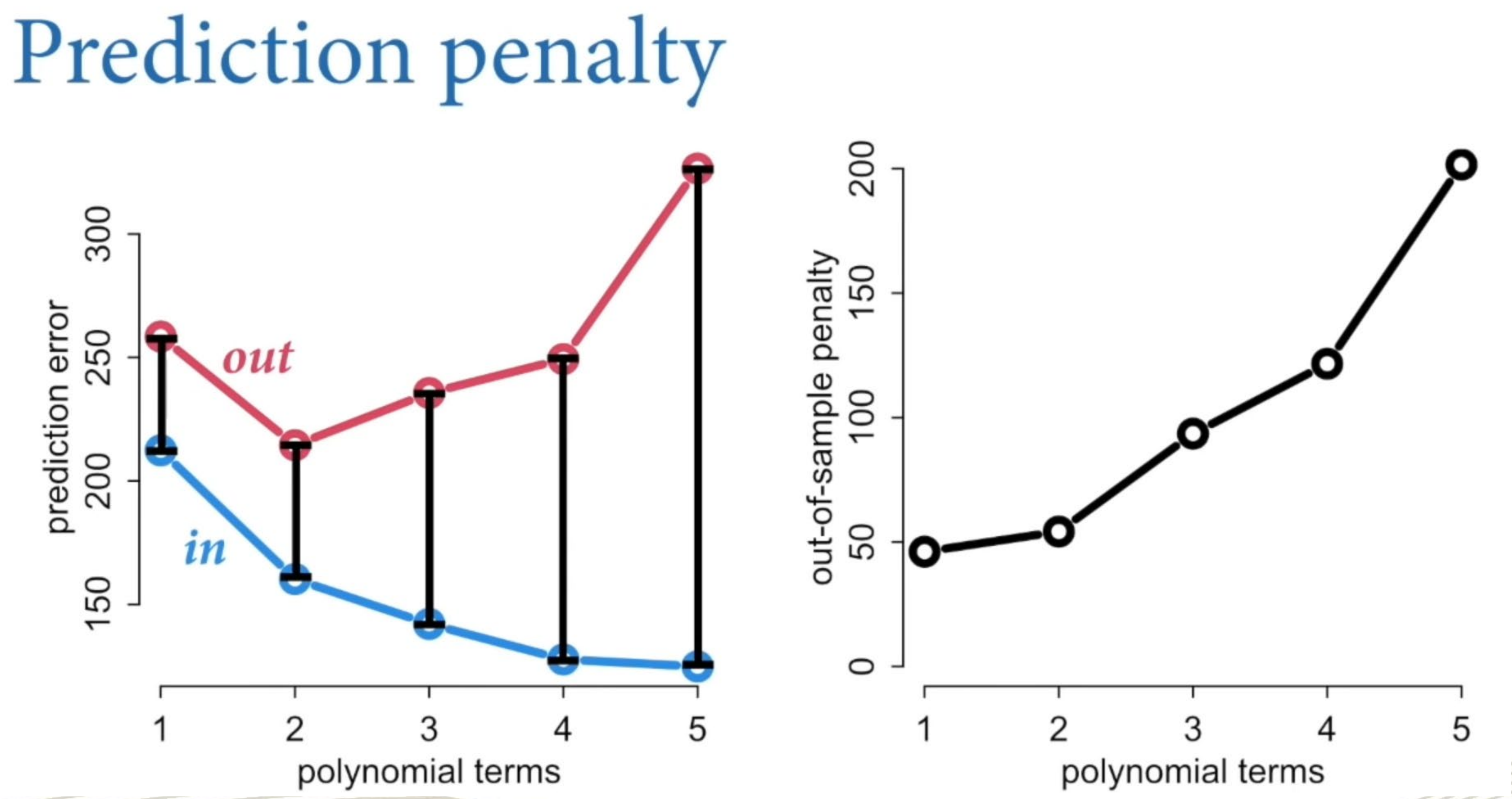
Information criteria

- Newer and more general approximations (of the out-of-sample deviance) exist that dominate AIC in every context: Widely Applicable Information Criterion (WAIC)

$$WAIC(y,\Theta) = -2(lppd - \underbrace{\sum_i \mathrm{var}_\theta \log p(y_i|\theta))}_{\text{Penalty term (effective number of parameters)}}$$

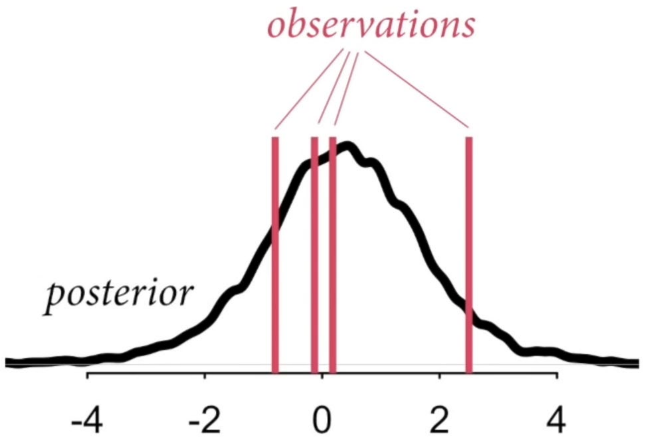


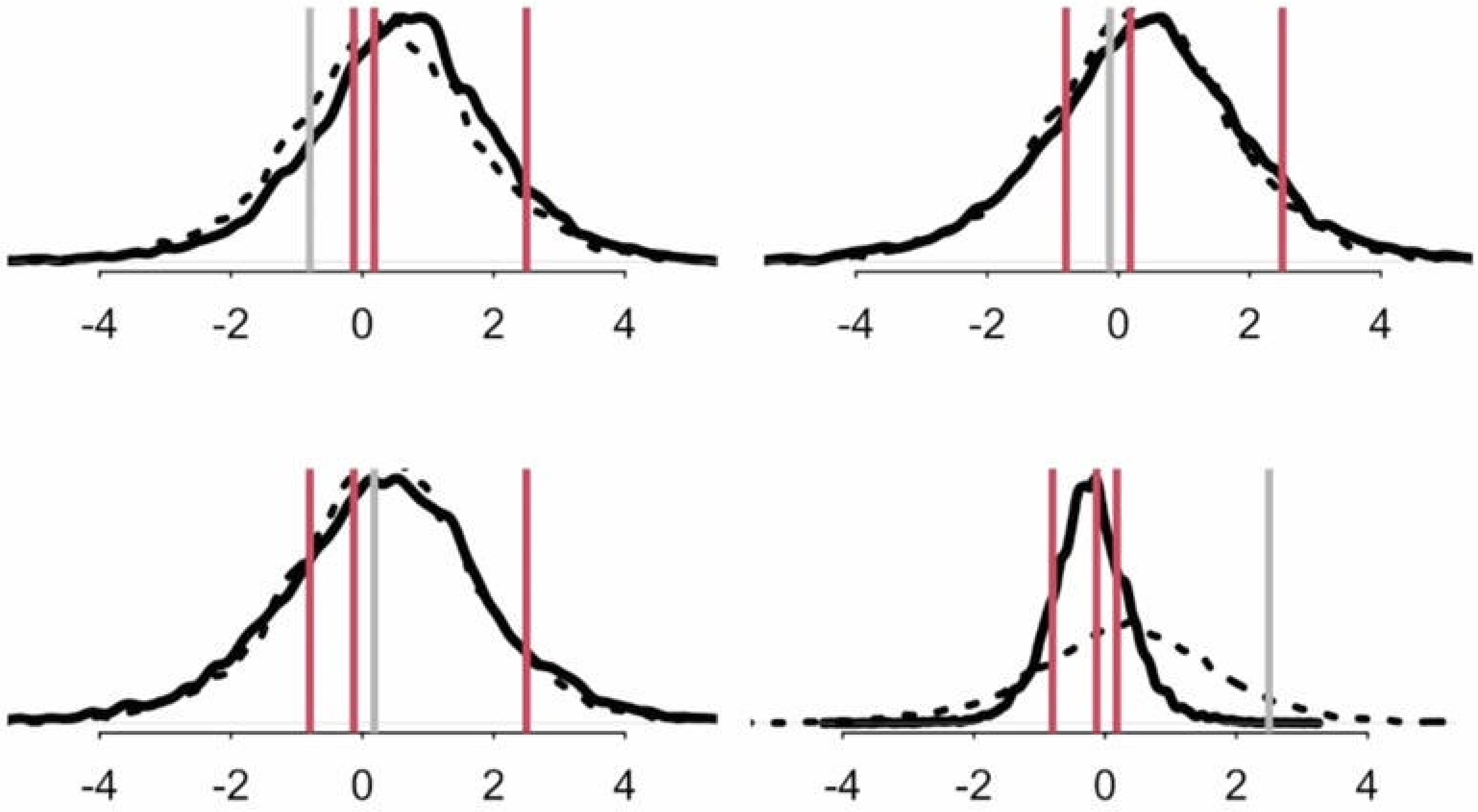
Prediction penalty



PSIS $p_M(y_i|y_{-i})$

Pareto smoothed importance sampling for leave-one-out cross-validation (LOO) approximation.





Podemos calcular la probabilidad posterior de dicha observación e inferir la posterior resultante sin la observación

PSIS $p_M(y_i|y_{-i})$

Pareto-smootheed (regularized) importance sampling cross-validation

Estimating out-of-sample pointwise predictive accuracy using posterior simulations

- Importance sampling replaces the computation of *N* posterior distributions by using an estimate of the importance of each i to the posterior distribution.

$$r_i^s = \frac{1}{p(y_i|\theta^s)} \propto \frac{p(\theta^s|y_{-i})}{p(\theta^s|y)}$$

Gelfand, A. E., Dey, D. K., and Chang, H. (1992).

PSIS $p_M(y_i|y_{-i})$

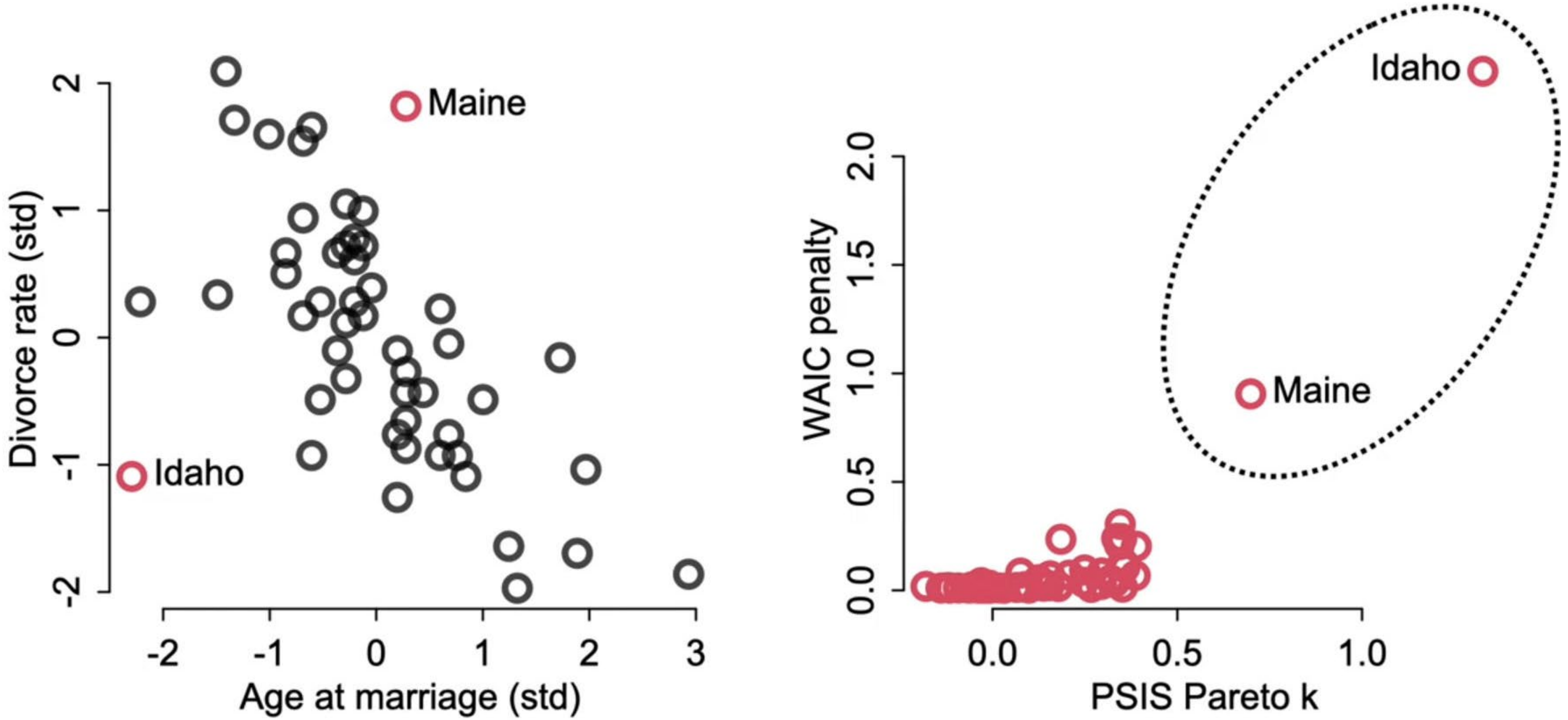
Pareto-smootheed (regularized) importance sampling cross-validation

- Re-weights each sample by the inverse of the probability of the omitted observation
  - Performed using existing simulation draws!
  - Also obtains approximate standard errors for estimated predictive errors and for comparison of predictive errors between two models.

$$r_i^s = \frac{1}{p(y_i|\theta^s)} \propto \frac{p(\theta^s|y_{-i})}{p(\theta^s|y)}$$

Gelfand, A. E., Dey, D. K., and Chang, H. (1992).

Quantify influence



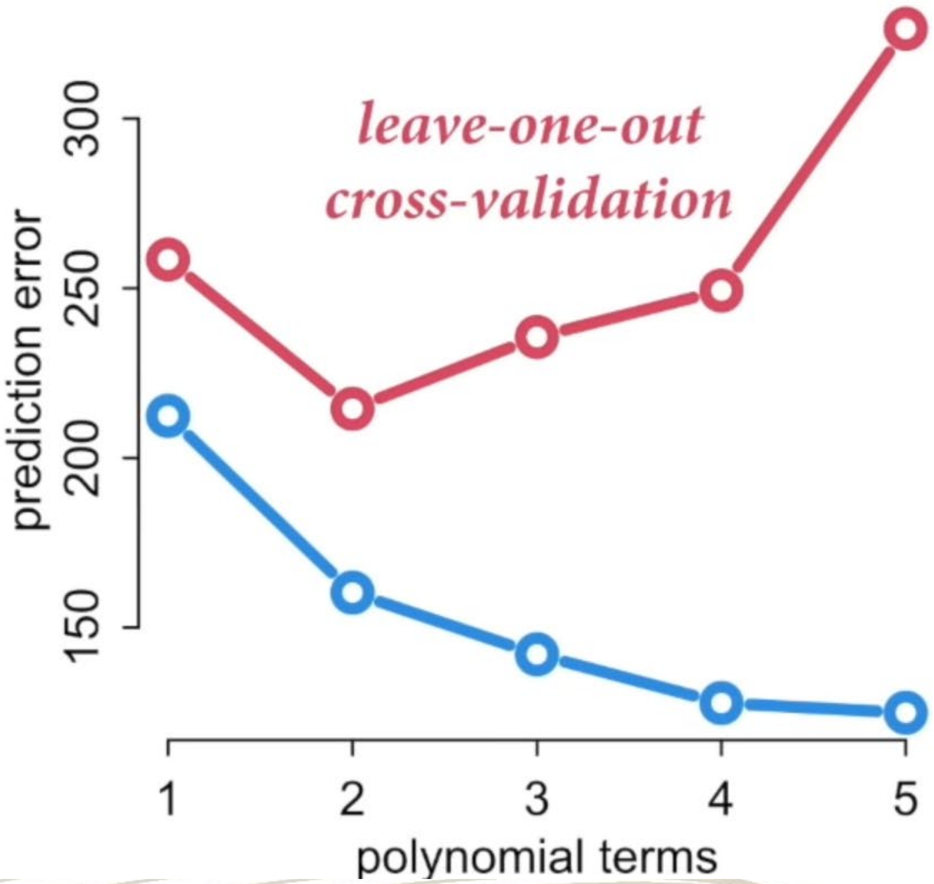
Observaciones en la cola de la distribución predictiva indican un posible exceso de confianza (el modelo no espera suficiente variación). Las predicciones no son confiables.

- PSIS: Much more stable and provides diagnostics
- Identifies outliers (indicate something is wrong with the model, dropping them only ignores the problem, predictions will still be bad)
- Estimate of the models performance out of sample

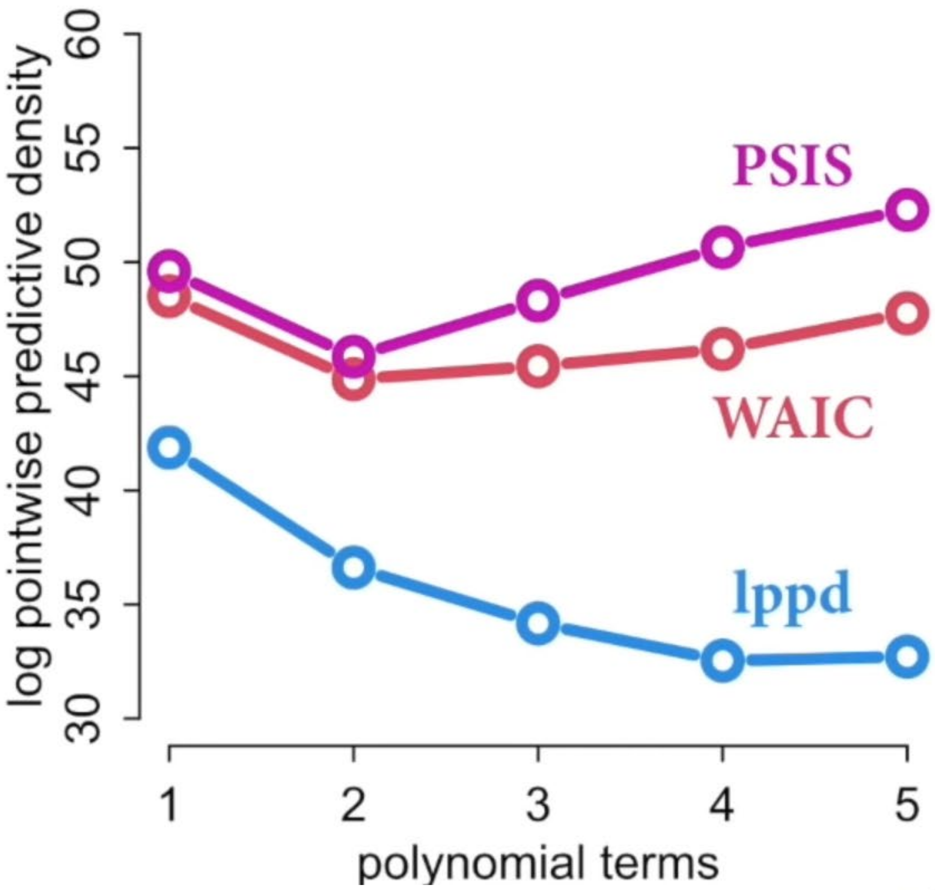
Prof Aki Vehtari (Helsinki), smooth estimator

Vehtari, A., Gelman, A., & Gabry J. (2016). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. In Statistics and Computing, doi:10.1007/s11222-016-9696-4. arXiv preprint arXiv:1507.04544.

Ejemplo



leave-one-out
cross-validation
prediction error
300
250
200
150
1
2
3
4
5
polynomial terms



PSIS
WAIC
lppd
log pointwise predictive density
60
55
50
45
40
35
30
1
2
3
4
5
polynomial terms

Estimación: efficient approximate loo

http://mc-stan.org/loo/reference/loo-glossary.html

```
loo1 <- loo(fit1, save_psis = TRUE)
print(loo1)
```

```
Computed from 4000 by 262 log-likelihood matrix

         Estimate     SE
elpd_loo  -6247.8  728.0
p_loo       292.4   73.3
looic     12495.5 1455.9
------
Monte Carlo SE of elpd_loo is NA.

Pareto k diagnostic values:
                         Count Pct.    Min. n_eff
(-Inf, 0.5]   (good)     239   91.2%   200
 (0.5, 0.7]   (ok)         6    2.3%   56
   (0.7, 1]   (bad)        8    3.1%   25
   (1, Inf)   (very bad)   9    3.4%   1
See help('pareto-k-diagnostic') for details.
```

The difference between elpd_loo and the non-cross-validated log posterior predictive density. It describes how much more difficult it is to predict future data than the observed data.

¡Cuando K pareto es grande!

$P\_loo > p$ (# parámetros): Mala especificación.

En casos bien comportados

$$p\_loo < N$$

y

$$p\_loo < p$$

Quisiéramos que todas fueran "buenas" <.7

Malas predicciones ☹

Lecciones

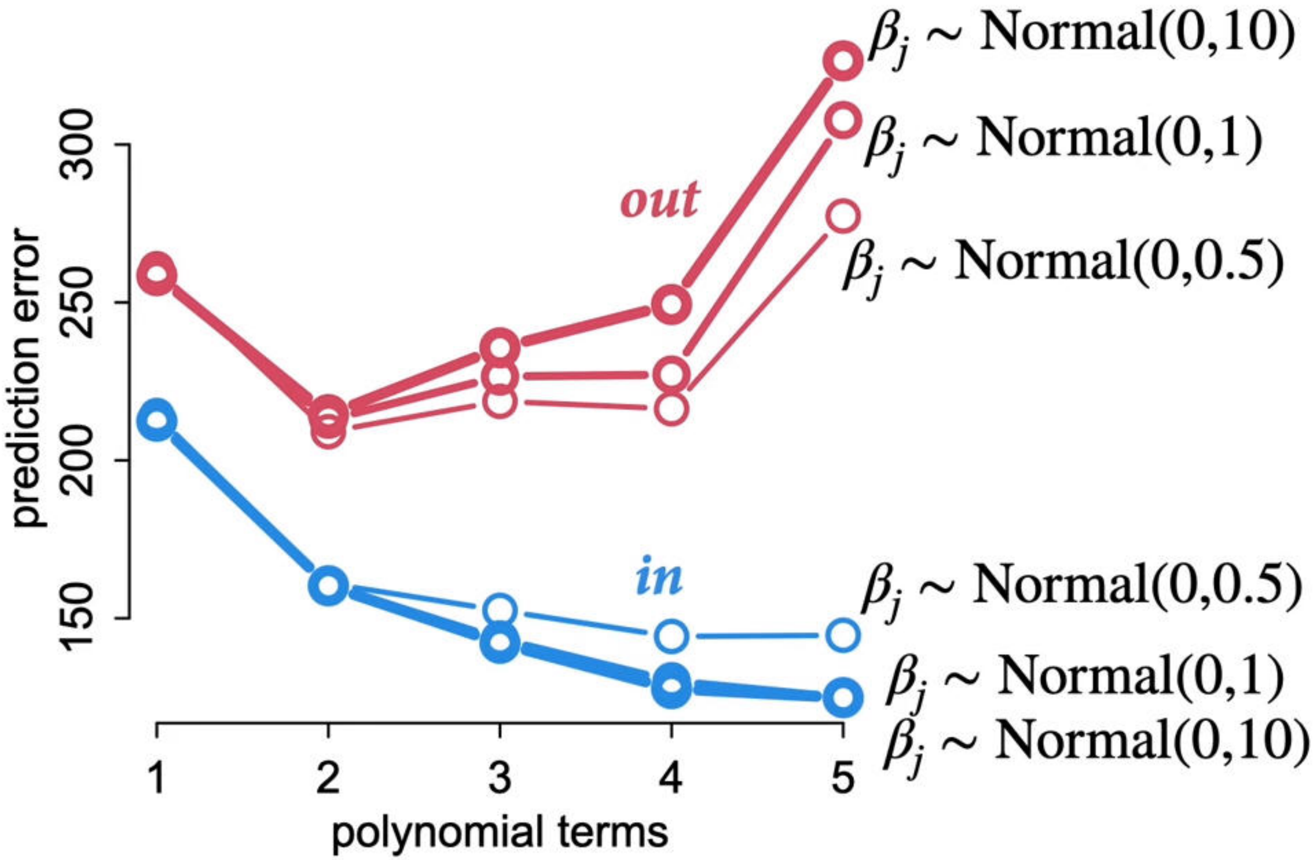
- NO USEN WAIC or PSIS para elegir modelos explicativos –causales-
- El modelo incorrecto puede ser mejor para predecir!

¿Y los priors?

- Cross-validation measures predictive accuracy, but does nothing about it.
- For pure prediction….Tune the prior using cross-validation (regularize, skeptical models tend to do better as they are less excitable)

Lecciones

- Flat priors are the worse
- You can make priors too tight/skeptical and learn nothing from the sample
- For causal inference… use science!
- Muchas tareas son una mezcla de inferencia y predicción



Referencias

Akaike, H. (1973), "Information theory and an extension of the maximum likelihood principle", in Petrov, B. N.; Csáki, F. (eds.), 2nd International Symposium on Information Theory, Tsahkadsor, Armenia, USSR, September 2-8, 1971, Budapest: Akadémiai Kiadó, pp. 267–281. Republished in Kotz, S.; Johnson, N. L., eds. (1992), Breakthroughs in Statistics, vol. I, Springer-Verlag, pp. 610–624

Akaike, H. (1985), "Prediction and entropy", in Atkinson, A. C.; Fienberg, S. E. (eds.), A Celebration of Statistics, Springer, pp. 1–24.

Ben-Naim, A. (2017). Entropy, Shannon's measure of information and Boltzmann's H-theorem. *Entropy*, *19*(2), 48.

Ben-Naim, A. (2008). *A Farewell to Entropy: Statistical Thermodynamics Based on Information: S*. World Scientific.

Ben-Naim, A. (2008). *Entropy demystified: The second law reduced to plain common sense*. World Scientific.

Celeux, G., & Soromenho, G. (1996). An entropy criterion for assessing the number of clusters in a mixture model. *Journal of classification*, *13*(2), 195-212.

Collins, L. M., & Lanza, S. T. (2009). *Latent class and latent transition analysis: With applications in the social, behavioral, and health sciences* (Vol. 718). John Wiley & Sons.

Gelfand, A. E., Dey, D. K., and Chang, H. (1992). Model determination using predictive distributions with implementation via sampling-based methods. In *Bayesian Statistics* 4, ed. J. M. Bernardo, J. O. Berger, A. P. Dawid, and A. F. M. Smith, 147{167. Oxford University Press.

Magnusson, M., Andersen, M. Jonasson, J., and Vehtari, A. (2019). Bayesian leave-one-outcross-validation for large data. *Proceedings of the 36th International Conference on Machine Learning*, 97:4244–4253.

Rioul, O., & LTCI, T. P. (2018, November). This is IT: A Primer on Shannon's Entropy and Information. L'Information, S´eminaire Poincar´e XXIII. http://www.bourbaphy.fr/rioul.pdf

Shannon, C. E. (1948). A mathematical theory of communication. *Bell system technical journal*, 27, July and October, 379-423 and 623-656.

Theil, H. (1967). *Economics and Information Theory*. Chicago: Rand McNally and Company.

Vehtari, A., Gelman, A., & Gabry J. (2016). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. In Statistics and Computing, doi:10.1007/s11222-016-9696-4. arXiv preprint arXiv:1507.04544.

https://www.youtube.com/watch?v=odGAAJDIgp8

CONTACTO

Dr. Héctor Nájera y Dr. Curtis Huffman
Investigadores (SNII II)

Programa Universitario de Estudios del Desarrollo (PUED)
Universidad Nacional Autónoma de México (UNAM)
Antigua Unidad de Posgrado (costado sur de la Torre II de Humanidades), planta baja.
Campus Central, Ciudad Universitaria, Ciudad de México, México.
Tel. (+52) 55 5623 0222, Ext. 82613 y 82616

Tel. (+52) 55 5622 0889
Email: hecatalan@hotmail.com, chuffman@unam.mx

¡Bienvenidos estudiantes!